Rupali Bhatnagar

Senior Software Engineer

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Work Experience

ServiceNow	Sr. Software Engineer (Security Operations)	Mar'22 – May'25

Individual contributor on Security Operations and Vulnerability Response, building on the ServiceNow© platform. Underlined features link to ServiceNow's public documentation.

- **Dynamic Approval Rule Engine** for vulnerability exceptions — designed a multi-level, dynamic approval rule engine on ServiceNow Flow Designer supporting deferment and exception requests across differing enterprise policies. **Reduced approval time by 40%.**
- **Penetration Testing Request Catalog & Workflow Automation** — built the catalog and automated the workflows for pen-testing requests, streamlining the vulnerability management lifecycle across **50+ subsystems** and **reducing manual intervention by more than 30%**.
- **Background Job Processing Framework** — architected a multithreaded framework for long-running tasks that previously strained system resources, with migration via a configuration interface:
  - Multithreaded execution — **60% efficacy improvement** in some use cases, **1M+ tasks/month**.
  - Job status tracking and cancellation — **100% visibility** on executing and errored tasks.
- **Real-Time Data Aggregation Engine** — high-performance aggregation framework for near real-time multi-dimensional analytics, powering key internal dashboards over **100K+ records in an 8-hour window**.

ServiceNow	Software Engineer (Security Operations)	Apr'19 – Mar'22

- **Assignment Rule Engine for Vulnerability Remediation** — cache-optimised engine prioritising infrastructure vulnerabilities and policy violations and routing them to the correct remediation groups via the Glide platform. Handles **1M+ vulnerable items daily**.
- **Machine Learning–Based Assignment Recommendations** — classification-based recommendation system with feature engineering and a user-facing review interface. **20% increase in task allocation accuracy.**
- **CISO & Vulnerability Dashboards** — led design and development of the CISO dashboard and supporting vulnerability dashboards, defining the key metrics and structuring the data pipeline. Used on a regular basis by **50+ organisations**.
- **Vulnerability Source Integration** — engineered integrations with **Veracode, Qualys, Tenable and Microsoft TVM**, standardising **500K+ vulnerability records/day** into a common model for downstream remediation workflows.
- **Hackathon Projects**:
  - **Diagnostic Framework** — streamlined issue tracking across customer instances, **reducing debugging time by 40%**.
  - **ML for Vulnerability Response** — models for grouping remediation tasks and estimating remediation targets.
  - **PoA&M Generator** — auto-generates Plans of Action & Milestones from vulnerability data; adopted by the NowOnNow support team.
  - **High-Volume Data Generator** — scalable generation of **100M+ records across 20–30 tables** for load testing; used by the performance team.

Darwinbox Digital Solutions	Software Developer	Aug'18 – Mar'19

- Designed and implemented a microservice architecture capturing **audit logs** for all client data modifications, written in **PHP with the Yii framework**, listening to **MongoDB oplogs** and publishing to an **AWS SQS** queue.
- Built an **Oplog Processor in Node.js** resolving differences across strings, arrays and embedded documents into human-readable audit entries, with an internal dashboard for adding trackable fields.

NowFloats (Boost360)	Software Developer	Jun'17 – Aug'18

- Developed responsive web features in **C#, Bootstrap, JavaScript, jQuery and Knockout.js**, supporting **50,000+ daily active users**.
- Integrated the **Facebook Graph API** so merchants could publish products and updates to their Facebook profile, page and shop (Ads extension).
- Built the **SPIDER** module using **machine learning** to predict sales revenue per representative from historical performance, with up to **84% accuracy**.
- Designed a **generic Booking System** for hotels, clinics and restaurants using **Node.js** and the WebActions API, and an **Order Management** dashboard (**AngularJS, C#**) for purchases, payment and delivery status and grievance redressal.

Finomena	Software Developer	Feb'17 – May'17

- Built internal dashboards for **Loan Management, Repayments, Collections and Transaction Tracking** in **PHP (Laravel)**, scaling to **8,000+ loans/month** with Eloquent ORM over MySQL.
- Used **Artisan** for task scheduling, cron handling and testing; Laravel modules for authorisation, validation, event handling and mailing.

Babajob.com	Software Engineer	May'16 – Jan'17

- Redesigned the **telephony system (IVR and SMS)** to scale to **10,000+ inbound calls and 20,000+ outbound SMS per day**, built in **Golang** with **RabbitMQ** as the message bus, **Redis** as in-memory cache and **MongoDB** for persistence.
- Designed and implemented RESTful outbound SMS services and a basic logging framework in Golang for debugging and tracing.
- Contributed to the **search subsystem in Java over Elasticsearch**, with dynamic query generation via Mustache templates.

AI & Agent Engineering

Tathya Dev — independent	AI & Agent Tooling	2025 – present

Agent-facing tooling and production LLM systems. Every item below is my own repository, running now:

- **Agent developer toolchain, deployed system-wide — propagate**, a Claude Code plugin published through a marketplace I run and installed at user level, so its hooks execute in **every agent session on the machine** rather than only where it is checked out. 4 slash commands, **4 hook scripts across 5 registrations on 2 events**, 63 library modules, **143 test files**. Currently watching **1,741 declared file couplings across 14 workspaces** against a **2,060-event** append-only store that reconciles in **~1.1s**. The marketplace serves **19 skills across 3 plugins**.
- **Agent guardrail delivered at the moment of risk** — a `PreToolUse` hook that pattern-matches the literal command an agent is about to execute against declared hazards and surfaces the matching one *before* it runs; written after a documented hazard sitting three files away was hit anyway and cost **11 false verification records**. A companion `SessionStart` hook injects **18 canonical engineering rules** into every session, and a restatement detector holds **50 `CLAUDE.md` files at zero divergent copies**.
- **Production LLM classification pipeline, engineered for unreliability** — Gemini calls constrained by enforced response schemas, run at `temperature: 0` and batched, with determinism resting on cache-on-first-write rather than on the model, and a keyword-classifier fallback so a provider outage degrades quality instead of breaking ingestion. Model-proposed sources are **existence-checked against the live API before acceptance**, and an inconclusive check returns the candidate flagged rather than asserted.
- **Evaluation harnesses, not vibes** — every detector and guardrail trigger ships a `selftest` asserting it can fire against a known-positive input, because **a check that cannot fail reports success**; the harness has caught real defects, including a rule whose detector could never have matched its own body. A Claude Cowork job-classification skill is tested against fixtures and a **golden set**.
- **Human-in-the-loop by construction** — no downstream file is ever edited automatically: the tooling finds and explains, a person approves, and each decision is recorded as an append-only ledger event. The background monitor has run **2,420 times with zero runs that could not look** — "found nothing" and "looked at nothing" are deliberately different outputs.

Selected Projects — 2025 to present

Independent and consulting engineering through **Tathya Dev** (tathya.dev).

Multi-tenant CRM & Workflow Platform Shree Shyam Jyotish Kendra (Pt. Pawan Kaushik) · ssjk-crm.vercel.app	Express API plus two Next.js 16 applications (client and admin). Built a **database-driven RBAC/ACL engine** — runtime-editable roles, generic field-level access control over CRUD, per-actor tier projection, table→field cascade, and a capability-matrix UI whose review-diff modal shows the consequences of a change before an atomic batch save, with fail-closed reconciliation and admin-floor guards. WebAuthn + OTP authentication, BullMQ/Redis job queue, workflow engine, web-push. Phased task-model migration dark-shipped in five stages with a one-shot backfill of **~14,000 records**. Test infrastructure cut from **78s to 28s** by running one shared `mongod` for the suite instead of 89. **228 test files.**
Content & SEO Platform Vipin Kaushik · vipinkaushik.com	Production Next.js site backed by the computation service below — the two are one system. Ten templated landing pages on a shared shell with **FAQPage structured data** and selective `noindex`, a server-side analytics pipeline, signed-cookie location handling with a durable resolved-location cache, and a migration of public pages to **static generation**. Largest single build of the last quarter.
Demand-Intelligence Platform Next.js, TypeScript, Mongoose, Gemini	Multi-source ingestion and opportunity scoring over Trends, YouTube and Search Console, fed by the schema-constrained Gemini classification pipeline described above, with an entity ontology mapping demand against supply. Search Console ingestion authenticated via **Workload Identity Federation**, adopted to remove service-account key fragility; scheduling moved off hosted cron onto a local agent.
Ephemeris Computation Service Python 3.12, FastAPI, MongoDB	FastAPI service over Swiss Ephemeris, a native library with process-global mutable state that is **not thread-safe**. Every call is serialised through a single-worker executor that releases the event loop during computation. Adds an astronomical-timeline subsystem (boundary and instant finders, lunar-month tables, per-year caching with TTL) and typed per-calculation endpoints replacing one overloaded route. **1,000+ tests.**
Motherboard Go 1.25, Next.js, Docker	Go multi-module monorepo implementing an API gateway and multi-tenant plugin platform — **25 `go.work` modules, 698 Go files**. Plugin services authenticate, publish their routes to a client manager, and the gateway proxies to them behind a **fail-closed entitlement gate** and a plugin-key claim assertion.
E-commerce Storefront & Embedded Admin Gemastrology · Shopify, Liquid, Remix, Prisma	Shopify Liquid theme plus an embedded Remix admin application — metafield/metaobject schema installer, headless catalogue seeders, and certificate verification implemented natively in Liquid rather than through an app proxy. Accessibility pass to a **44×44 touch-target floor**.
Production Portfolio Sites Manav Daehi · Kirti Kumar · Youvan Prakashan	manavdaehi.com — Google Docs→works sync pipeline and multi-codec delivery (AV1/HEVC/H.264); a hero codec ladder took **10.4 MB to ~3 MB**, with delivery suppressed for non-watchers and an H.264 Level 4.0 re-encode for iOS autoplay. kirtik.co.uk — dual-track portfolio with Playwright end-to-end coverage. youvanprakashan.org — publishing platform with an Express admin classification queue and a language classifier.
Text Digitisation & Translation Pipeline Python · github.com/Rupali59/Sanskrit-texts	Ingestion for a structured-JSON corpus of classical texts: a TEI/SARIT parser handling prose and division-type detection, a scraper splitting on the source's own headings, a verse-marker grammar, and per-work structural validators. One structural fix took depth errors in a single work from **3,522 to 6**. Corpus spans **90,000+ verses across 50+ texts**.

Dissertation & Coursework

Dissertation	Human Activity Recognition using an **HMM-based Intermediate Matching Kernel**, representing videos as sequences of sets of feature vectors. Re-implemented in 2026 as a containerised Python pipeline with a Flask front end — github.com/Rupali59/himk-implementation.
Methodology	Classifies a video into a predefined activity class. Feature extraction and classification in Python with scikit-learn, hmmlearn and OpenCV; originally prototyped in MATLAB with Piotr's CV Toolbox, PNetCDF and OpenMPI.
ML Techniques	Bayesian classifiers, Gaussian mixture models, perceptrons, artificial neural networks and support vector machines on linearly separable, non-linearly separable and overlapping data.
Other Projects	Management Information System (MIS), Cybernetics Protector System, Product Management System.

Technical Skills

Languages	Go, TypeScript, JavaScript, Python, PHP, Java, C#, C++, GlidePlatform
Web & Frameworks	Next.js, React, Node.js, Express, Remix, Astro, Laravel, .NET, FastAPI, Tailwind, Bootstrap
AI & Agent Tooling	Claude Code plugins, hooks and skills; structured LLM output and schema-constrained generation; eval harnesses and golden sets; hallucination containment and graceful degradation; human-in-the-loop design; MCP
Databases	MongoDB, PostgreSQL, MySQL, Redis, Elasticsearch
Queues & Messaging	RabbitMQ, BullMQ, AWS SQS, Redis Queue
Cloud & DevOps	AWS, Vercel, Fly.io, Docker, Shopify, Jenkins, Git, Linux
ML	scikit-learn, hmmlearn, OpenCV, pattern recognition and classification

Academic Profile

Year	Degree	Discipline	Institution	Score
2016	**M.Tech**	Computer Science & Engineering Pattern Recognition & Machine Learning	National Institute of Technology, Goa	8.42 CGPA
2014	**B.E.**	Computer Science & Engineering	LNCT, Bhopal (RGPV)	8.58 CGPA
2010	XII (CBSE)	Science	St. Joseph's Convent School, Bhopal	83.0%
2008	X (CBSE)	—	St. Joseph's Convent School, Bhopal	88.0%

Certifications

AI — 2026	**Anthropic:** Claude Code in Action; Introduction to Claude Cowork; Claude Platform 101; Claude Code 101; Claude 101. **Google:** Gemini Enterprise Agent Ready; Create Your First Gemini Enterprise Application; Google Cloud Innovator; Developer Program Premium Tier; AI Fundamentals (Google Career Certificates). **Verify:** credly.com/users/rupali-bhatnagar-59 · g.dev/rupali59.
Earlier	WhatsApp API Expert, Wati (2024) · Ultimate JavaScript: Fundamentals, Code With Mosh (2025) · Neural Networks & Deep Learning, Coursera (2020) · JavaScript Algorithms & Data Structures, freeCodeCamp (2020) · Git Essential Training, LinkedIn Learning (2020)

Achievements & Activities

Professional	• Winner, NowFloats Hackathon for product innovation in sales — built the ML-powered SPIDER revenue-prediction module. 2018
Academic	• **Chancellor's Scholarship (Topper)**, Rajiv Gandhi Proudyogiki Vishwavidyalaya, Bhopal — awarded to the top 10 students statewide by theory marks. 2012, 2013 • **Silver Medal**, Persistent Systems Limited Programming Contest, inter-college. 2013
Other	• Organised the intra-college programming competition, LNCT Bhopal. 2012 • Conducted cultural events, NIT Goa. 2014–2016

Positions of Responsibility

NIT Goa	Teaching Assistant, Computer Science & Engineering Department	2014–2016
NIT Goa	Vice President, Computer Science Department	2014–2016
LNCT, Bhopal	Student Coordinator, Training & Placement Department	2013–2014